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Incidence of metabolic syndrome according to combinations of lifestyle factors among middle-aged Japanese male workers

Yuanying Li [a,b], Hiroshi Yatsuya [b,c,*], Hiroyasu Iso [a], Koji Tamakoshi [d], Hideaki Toyoshima [e]

[a] Public Health, Department of Social and Environmental Medicine, Osaka University Graduate School of Medicine, Osaka, Japan
[b] Department of Public Health/Health Information Dynamics, Nagoya University Graduate School of Medicine, Nagoya, Japan
[c] Division of Epidemiology and Community Health, School of Public Health, University of Minnesota, Minneapolis, MN, USA
[d] Department of Nursing, Nagoya University School of Health Sciences, Nagoya, Japan
[e] Health Care Center, Anjo Kosei Hospital, Anjo, Aichi, Japan

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ABSTRACT

Objective. To examine a combination of healthy lifestyles on the risk of metabolic syndrome (MetS) to inform future interventions.

Methods. A total of 1897 men aged 35–60 years participated in an annual health check-up in 2002 and 2005. MetS was defined by AHA/NCEP criteria. Logistic regression analyses were used to estimate age- and BMI-adjusted odds ratios (ORs) of MetS incidence for each healthy lifestyle (regular physical activity, adherence to healthy eating behaviors, not current smoking, and maintaining a stable weight since one's mid-twenties), separately (Model 1) and simultaneously (Model 2). A points system was developed to derive 3-year risk of MetS incidence by assigning a specific point to each healthy lifestyle.

Results. MetS developed in 285 (15.0%) subjects after the follow-up. The ORs of MetS for each healthy lifestyle ranged from 0.42 to 0.64 (Model 2). Three-year risk of MetS incidence was predicted to differ from 1% to 60% according to the individual point total of the points system. The population-attributable fraction of MetS in subjects whose point total was not in the highest quartile was 71%.

Conclusion. Adherence to healthy lifestyles was associated with a lower risk of MetS among apparently healthy middle-aged Japanese male workers.

Introduction

Metabolic syndrome (MetS) is a high-risk state for the development of cardiovascular diseases (CVD) (Iso et al., 2007; Ballantyne et al., 2008) and diabetes mellitus (Klein et al., 2002), even though the degree of abnormality in each clustered component is mild to moderate. The concept has been challenged regarding whether or not more efficient interventions can be addressed to prevent CVD and diabetes than just targeting individual components (Morabia and Costanza, 2006). On the other hand, previous studies have investigated the association of modifiable lifestyle factors, for example, dietary pattern (Lutsey et al., 2008; Esmaillzadeh et al., 2007; Esmaillzadeh and Azadbakht, 2008), smoking, alcohol consumption, physical activity (Park et al., 2003; Zhu et al., 2004), and weight change (Zhu et al., 2004; Zhang et al., 2005) with the risk of MetS separately. Because lifestyle factors are inter-correlated, and there may be complex interactions between them, it is important to investigate their combined effects on the incidence of MetS so as to inform future interventions such as multiple health behavior change intervention (Prochaska, 2008; Morabia and Costanza, 2008). But so far, no study has investigated the combined effects of lifestyles on the risk of MetS prospectively. We, therefore, examined the combination of lifestyle factors such as physical activity, eating behaviors, smoking status, and weight change since one's mid-twenties in relation to the risk of MetS incidence in a cohort of middle-aged Japanese male workers.

Methods

Study population

A baseline survey was conducted in 2002, and data were available on self-reported medical history and lifestyle factors as well as the result of an annual health check-up for 4317 men and 1140 women with their written consent. Because of the small number of women who developed MetS during follow-up ($n = 9$), we limited the present analysis to men. We excluded those who had a history of cancer or CVD at baseline ($n = 85$), had been on medical treatment for hypertension, hyperlipidemia, or diabetes ($n = 465$), or had implausibly low or high estimated caloric intake (<500 or ≥3500 kcal per day) ($n = 66$). We further excluded those with MetS at baseline ($n = 655$). Among the remainder, 1897 subjects who had undergone an annual health

* Corresponding author. Department of Public Health/Health Information Dynamics, Nagoya University Graduate School of Medicine, 65 Tsurumai-cho, Showa-ku, Nagoya 466-8650, Japan. Fax: +81 52 744 2131.
E-mail address: yatsuya@gmail.com (H. Yatsuya).

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check-up in 2005 and with no missing information for the diagnosis of MetS incidence were left for the present analysis. The majority of men who did not attend the follow-up survey had retired. Those who did not undergo the 2005 health check-up were 3 years older than those who did. The ethics Review Board of Nagoya University School of Medicine approved the study protocol.

Healthy lifestyles

Four healthy lifestyles were selected. First, adherence to healthy eating behaviors was evaluated by summing the number of eating behaviors each subject had at baseline: not preferring salty foods (Hori et al., 2003), eating breakfast almost daily (van der Heijden et al., 2007), not eating fast (Otsuka et al., 2006; Maruyama et al., 2008; Otsuka et al., 2008), and not eating until full (Maruyama et al., 2008; Toyoshima et al, 2009). Subjects with all four healthy eating behaviors were considered as having a healthy lifestyle factor. Second, subjects who quit smoking as well as those who have never smoked (Chen et al., 2008; Oh et al., 2005) were considered to have a healthy lifestyle in terms of smoking status. Third, regarding physical activity, persons who engaged in moderate or vigorous activity 3 days or more and 90 min or more in total per week (Laaksonen et al., 2002; Park et al., 2003) were considered as regularly physically active, and having a healthy lifestyle. Fourth, weight change was calculated as the difference between recalled weight at about mid-twenties (Tamakoshi et al., 2003) and measured baseline weight, and subjects with weight change within ± 3 kg were considered here as maintaining a stable weight as a surrogate marker for overall healthy lifestyles (Field et al., 2004).

Dietary pattern and alcohol intake

We initially considered dietary pattern and alcohol drinking habit, respectively, as one of the healthy lifestyles. Healthy dietary patterns were defined by a priori classification of healthy food items as previously published (Sarrafzadegan et al., 2009) as well as by a factor analysis that yielded the vegetable and fruit dietary pattern. However, neither approaches revealed its significant association with MetS incidence independent of other healthy lifestyles. Similarly, neither light nor no drinking was associated with the incidence of MetS independently. Thus, we did not take them into account in the calculation by the points system, and did not present their results.

Anthropometric measurements and biochemical analysis

Weight and height were measured with the subjects in typical indoor clothing but without shoes, weight to the nearest 0.5 kg, and height to the nearest 0.1 cm. Body Mass Index (BMI) was calculated as weight (in kg) divided by the square of height (in meter). Waist circumference was measured to the nearest cm at the umbilical level in the late exhalation phase while standing. Blood pressure was measured for least 5 min of rest using an automatic sphygmomanometer (BP-103N II; Omron-Colin Co, Tokyo, Japan). Venous blood samples were drawn after the subjects fasted for 8 h or overnight, and serum samples were stored at −80 °C until the biochemical assay. Fasting blood glucose was enzymatically determined by the hexokinase method. Triglycerides and high-density lipoprotein cholesterol were measured enzymatically and by the phosphotungstate method, respectively.

Definition of MetS

We used the minimally modified American Heart Association/National Heart, Lung, and Blood Institute (AHA/NHLBI) criteria for MetS (Grundy et al., 2005). Those with three or more of the following abnormalities at follow-up were considered as having MetS incidence. Abdominal obesity was defined as waist circumference of 85 cm or greater according to the criteria of the Japan Society for the Study of Obesity (Examination Committee of Criteria for 'Obesity Disease' in Japan, 2002). At baseline, BMI 25 kg/m^2 or greater was used instead since waist circumference was not available. High blood pressure was defined either by systolic blood pressure of 130 mm Hg or greater or diastolic blood pressure of 85 mm Hg or greater. High fasting glucose was defined as blood glucose of 100 mg/dl (5.55 mmol/l) or greater; high triglycerides was defined by 150 mg/dl (1.70 mmol/l) or greater; and low high-density lipoprotein cholesterol less than 40 mg/dl (1.04 mmol/l).

Statistical analysis

Differences in baseline characteristics were tested by chi-squared test and one-way analysis of variance (ANOVA). Baseline age- and BMI-adjusted binominal logistic regression analyses were used to estimate odds ratios (ORs) and 95% confidence intervals (CIs) of MetS incidence for each healthy lifestyle factor, separately (Model 1) and simultaneously (Model 2). In dichotomizing subjects into healthy and non-healthy, those whose responses in a given healthy lifestyle were missing were considered here to be in a non-healthy category. No subjects had more than two missing variables out of four healthy lifestyles.

We developed a points system according to the method used in the Framingham Study (Sullivan et al., 2004). The point assigned for the presence of each healthy lifestyle was derived from the regression coefficients (β) of the multivariate model (Model 2), which essentially contains weight for the healthy lifestyle. The regression coefficients were estimated in a model that included all the four healthy lifestyles (yes, no), age (continuous) and BMI (continuous). We derived points of age and BMI in categories (35–49, 50–60 for age) and (<21, 21–<23, 23–<25, 25–<27, 27 for BMI). A point total was calculated for each individual, and participants were categorized into quartile based on the distribution of the predicted 3-year risk according to the point total. Quartile-specific 3-year risk was re-calculated as a weighted average of risks predicted for each point total included in the quartile (0–7, 8–9, 10–12, and 13–22). Risk ratio was derived by dividing the estimated 3-year risk for a quartile being considered by the one for the lowest risk quartile. To estimate the proportion of subjects who would theoretically not have developed MetS if they had adhered to the healthier lifestyles, the population-attributable fraction (PAFs), was calculated as $P \times [(\text{risk ratio for a quartile being considered} - 1) / \text{risk ratio for a quartile being considered}]$ where P is the proportion of MetS cases exposed in the quartile being considered (Greenland, 1999).

To evaluate discriminatory ability of the point system, area under the curve (AUC) of receiver operating characteristic curve was estimated. We also compared the point system AUC with that of a model using original logistic regression coefficients to test possible loss of information.

All statistical analyses were conducted with the SPSS statistical package for Windows version 16.0 (SPSS Inc.).

Results

Mean age of the subjects was 46.4 years, and mean BMI was 22.6 kg/m^2 at baseline (Table 1). MetS developed in 15.0% ($n = 285$) of the subjects after 3 years of follow-up, and abdominal obesity was the most common disorder (44.4%) among the MetS cases, followed by hypertension (42.6%), high triglycerides (21.4%), elevated fasting blood glucose (19.9%), and low high-density lipoprotein cholesterol (5.8%).

Each healthy lifestyle was inversely associated with MetS incidence with the ORs ranging from 0.41 to 0.63 (Model 1 in Table 2). The ORs in Model 2 simultaneously adjusted all the healthy lifestyles did not materially alter.

Points assigned to healthy lifestyles as well as each category of age and BMI, are presented in Table 3. Regular physical activity had the largest positive point (+4) followed by adherence to healthy eating behaviors (+3) among the four lifestyles. The number of MetS cases, the estimated 3-year risk of MetS and PAFs according to the point total are shown in Table 4. The PAF of MetS for having a point total of 12 or less was 71% compared with having a point total of 13 or more.

Table 1
Mean (standard deviation) age, body mass index and each metabolic syndrome component of the study sample at baseline, Aichi, Japan, 2002.

Age (year)	46.4 (6.1)
Body mass index (kg/m$^{2)}$	22.6 (2.3)
Systolic blood pressure (mm Hg)	123.9 (13.2)
Diastolic blood pressure (mm Hg)	76.6 (10.3)
Fasting blood glucose (mg/dl)	93.9 (14.8)
Triglycerides (mg/dl)	116.9 (75.0)
High-density lipoprotein cholesterol (mg/dl)	57.2 (14.8)

Table 2
Odds ratios (95% confidence intervals) of metabolic syndrome incidence in relation to healthy lifestyle factors, Aichi, Japan, 2002–2005.

Healthy lifestyles		Number of Subjects	Number of cases (%)	Odds ratio (95% CI)	
				Model 1[a]	Model 2[b]
Physical activity	Regular	149	11 (7.4)	1 (reference)	
	Not regular	1502	234 (15.6)	2.42 (1.27–4.60)	2.39 (1.25–4.57)
	Missing	246	40 (16.3)	2.49 (1.21–5.09)	2.41 (1.17–4.96)
	Regular vs. others	149 vs. 1,748	–	0.41 (0.22–0.78)	0.42 (0.22–0.80)
Number of healthy eating behaviors[c]	4	202	13 (6.4)	1 (reference)	
	3	487	60 (12.3)	1.76 (0.93–3.31)	1.70 (0.90–3.22)
	2	696	104 (14.9)	2.04 (1.11–3.75)	2.00 (1.08–3.69)
	≤1	512	108 (21.1)	2.68 (1.45–4.97)	2.50 (1.34–4.67)
	4 vs. ≤3	202 vs. 1,695	–	0.47 (0.26–0.85)	0.49 (0.27–0.89)
Smoking status	Never	740	94 (12.7)	1 (reference)	
	Former	488	64 (13.1)	0.93 (0.66–1.33)	0.90 (0.63–1.28)
	Current	660	125 (18.9)	1.61 (1.19–2.18)	1.54 (1.13–2.10)
	Missing	9	2 (22.2)	1.30 (0.26–6.52)	1.02 (0.19–5.50)
	Not current smoking vs. others	1,228 vs. 669	–	0.61 (0.46–0.79)	0.62 (0.48–0.82)
Weight change since one's mid-twenties (kg)	≥−27 and <−3	111	16 (14.4)	2.20 (1.19–4.07)	2.10 (1.13–3.91)
	Stable (≥−3 and ≤3)	644	53 (8.2)	1 (reference)	
	>3 and ≤9	642	98 (15.3)	1.45 (1.01–2.09)	1.42 (0.98–2.05)
	>9 and ≤30	398	99 (24.9)	1.68 (1.11–2.52)	1.64 (1.08–2.48)
	Missing	102	19 (18.6)	1.60 (0.88–2.89)	1.67 (0.90–3.08)
	Stable vs. others	644 vs. 1,253	-	0.63 (0.45–0.88)	0.64 (0.46–0.90)

[a] Model 1: adjusted for age (continuous) and body mass index (continuous).
[b] Model 2: adjusted for covariates in model 1 and other lifestyle factors.
[c] Healthy eating behaviors include not preferring salty foods, eating breakfast almost everyday, not eating fast and not eating until full.

The point system appeared to discriminate those who would and would not develop MetS well in the studied population with AUC of 0.72 (95% CI: 0.69–0.75). The AUC was materially the same as that by a model using original logistic regression coefficients (0.73, 95% CI: 0.70–0.76).

Discussion

In the present analysis, we found that the more healthy lifestyles the subjects had at baseline, the less likely their risk of developing MetS after 3-year follow-up. Although the points system incorporated not only lifestyles but also age and BMI, the majority (about 70%) of the MetS cases could be attributable to the poor adherence to some or all of the healthy lifestyles in this cohort of apparently healthy Japanese male workers. Supplementary analysis excluding age and BMI yielded virtually similar results (data not shown). These results are generally consistent with previous large cohort studies showing that the subjects adhering to five healthy lifestyles had less than one-tenth the risk of myocardial infarction (Akesson et al., 2007) and diabetes mellitus (Hu et al., 2001), persons with four healthy lifestyles had less than one-fifth the risk of coronary event (Stampfer et al., 2000) compared to persons with no healthy lifestyle.

Regarding each lifestyle, our findings were also in accordance with the previous literature on smoking (Chen et al., 2008; Oh et al., 2005), and physical activity (Laaksonen et al., 2002; Park et al., 2003).

The number of healthy eating behaviors, each of which was shown to be associated with MetS or its components, was inversely associated with risk of MetS (Hori et al., 2003; van der Heijden et al., 2007; Otsuka et al., 2006; Maruyama et al., 2008; Otsuka et al., 2008; Toyoshima et al., 2009). Furthermore, maintaining a stable weight since one's twenties was associated with a lower risk of MetS as shown in previous studies (Zhang et al., 2005; Yatsuya, 2007). Because weight change may not be viewed solely as a lifestyle factor, we conducted a sensitivity analysis excluding weight change variable from healthy lifestyle factors, but the result did not change materially (data not shown).

We initially attempted to incorporate healthy lifestyles regarding diet and alcohol consumption; however, both lifestyles did not have significant associations with MetS incidence in the present sample. Lack of association between diet–factor analysis derived food pattern high in vegetable and fruit or a priori-defined healthy diet (Sarrafzadegan et al., 2009)–might have arisen from the large variability of the diet in the present sample: the vegetable and fruit dietary pattern was the first-derived pattern in the factor analysis, which explained only 9.8% of the diet variance. In other populations, for example, the Western dietary pattern alone explained 19% of the dietary intake variance (Deshmukh-Taskar et al., 2009). The significant association of alcohol intake with MetS incidence disappeared after adjustment for other lifestyles. Although previous studies found its independent association with

Table 3
Points assigned for each healthy lifestyle factor, age and body mass index categories, Aichi, Japan, 2002–2005.

Healthy lifestyles, age and BMI categories	Categories	Points
Regular physical activity	Yes	4
	No	0
Adherence to four healthy eating behaviors	Yes	3
	No	0
Not current smoking	Yes	2
	No	0
Maintaining stable weight since mid-twenties	Yes	2
	No	0
Age (year)	35–49	3
	50–60	0
Body mass index (kg/m^2)	−21	8
	21–23	6
	23–25	4
	25–27	2
	27-	0

Table 4
Three-year risk of metabolic syndrome incidence and population-attributable fraction according to quartiles of point total, Aichi, Japan, 2002–2005.

Point total	N of subjects	N of MetS (%)	Three-year risk	PAFs (%)
13–22	485	16 (5.6)	0.05	Reference
10–12	442	41 (14.4)	0.10	7
8–9	445	76 (26.7)	0.16	19
0–7	525	152 (53.3)	0.30	45

N denotes number; MetS, metabolic syndrome; PAF, population-attributable fraction. Point total is calculated from Table 3.

MetS (Freiberg et al., 2004), the cardiometabolic effect of alcohol consumption is controversial (Yoon et al., 2004; Baik and Shin, 2008), which may partly support the present insignificant finding. These findings, however, would not diminish the importance of dietary factors as a determinant of MetS, but may reveal some difficulties in their assessment. In addition, some nutrients such as trans fatty acids were not evaluated in the present study (Micha and Mozaffarian, 2009). Future investigations incorporating the dietary pattern and specific nutrients would be useful.

The derived points system appeared able to distinguish subjects with low and high risk of MetS incidence in 3 years. According to this system, regular physical activity and healthy eating behaviors were assigned the highest points (+4 and +3, respectively) among the four lifestyles, indicating that by addressing these two lifestyles only (+7 points change together), an individual could be transposed from the lowest quartile of the point total to the highest quartile, which would reduce the individual risk of MetS substantially, and it would also eliminate a large number of MetS cases in the population. However, the point system itself should be validated when it is applied to other populations. We did not include precision of the estimated risk in the point system, either. Further research addressing these issues would be needed before implementing it into intervention.

Since some healthy lifestyles were related with one another, one may assume that those who reported multiple healthy lifestyles selected in the present study were likely to have other unmeasured or unknown healthy lifestyles, which may have contributed to lowering the risk of MetS. It is also possible that several coexisting lifestyles may have synergistic effects. One recent intervention study reported that subjects assigned exercise intervention together with a high protein diet acquired a better metabolic profile than those on a high carbohydrate diet (Layman et al., 2005). Thus, an intervention study would be required to confirm whether achieving these four healthy lifestyle factors would indeed reduce the risk of MetS to the degree observed in the present study, or to correctly evaluate the complexity of a specified combination of multiple healthy lifestyles.

The magnitude of risk reduction related to having a point total of 13 or more was substantial (six-fold risk reduction compared with a point total of 7 or less) in our analysis. However, the points system itself should be evaluated elsewhere with precisely standardized and easily applicable measurement of lifestyles in order to generalize to other populations.

Some limitations warrant consideration. First, healthy lifestyles selected and defined here are rather arbitrary, and the prevalence of the healthy lifestyles as well as their association with MetS would differ by population. Thus, although the large number of MetS in the present sample could be prevented by these lifestyles, the observed results can not be readily generalized so that the majority of MetS can be eliminated by changes in these lifestyle factors. The present finding should be confirmed in other populations with a different age, sex and ethnicities, and including subjects residing in rural communities or blue-collar workers. Another limitation is the short time of follow-up, i.e., 3 years. Whether adhering to these healthy lifestyles will persist and result in a long-term reduction of MetS risk would need further investigation. In addition, the number of MetS incidence in individuals with high points (16 or more) was very small. Longer follow-up would be needed to confirm the magnitude of the association. Misclassification due to change in the lifestyle during follow-up especially in response to a variable that is associated with outcome variable (e.g., weight gain) might possibly have distorted the healthy lifestyle and MetS association. This misclassification might explain the insignificant associations between dietary pattern and alcohol intake and MetS incidence. Further analysis incorporating the change in the exposure variable may be needed especially with the extended follow-up. Finally, the number of missing responses was moderate. We considered subjects with missing responses in a given healthy lifestyle to be in a non-healthy lifestyle category to yield the most conservative estimate of the risk. In addition, although subjects with missing information were 1.1 years older than those without, there were no differences in age-adjusted MetS incidence. Restricting analysis to men without any missing values ($n = 1528$) did not appreciably change the present results (data not shown).

The strength of our study included a prospective population-based design and the availability of information about a variety of lifestyles. By simultaneous evaluation of these lifestyles, we observed the combined effect of lifestyles on risk of MetS.

In summary, healthy lifestyle factors were inversely associated with risk of MetS among middle-aged Japanese male workers. The majority of MetS could be prevented by adherence to those healthy lifestyles. To confirm the potential of lifestyle modification, further observational studies in different populations and intervention studies are warranted.

Conflict of interest statement

The authors declare that there are no conflicts of interest.

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